

HACKBACK · WORKFLOW & TRUST · KING OF HEARTS

Smart Slot Booking

Reverse-engineering Cal.com

200 students need to book faculty office hours and lab slots across time zones, with buffers and day-off overrides, and no two people in the same slot.

Team DBG-922

The original: Cal.com

LANGUAGE

TypeScript

FRAMEWORK

Next.js + React

API

tRPC + REST

DATABASE

Postgres +
Prisma

MONOREPO

Turborepo + Yarn

AUTH

NextAuth.js

STYLING

Tailwind CSS

DATES

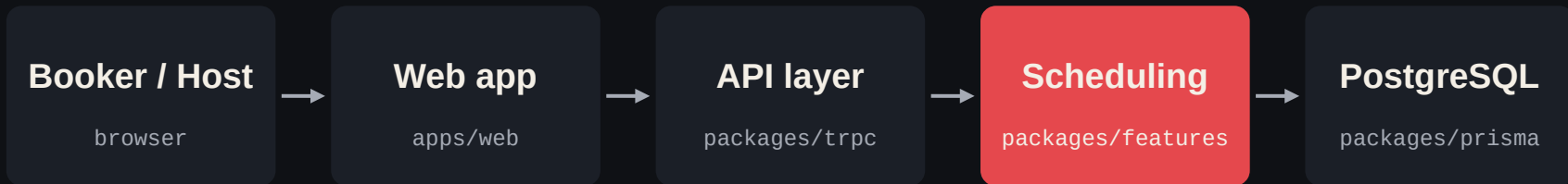
Day.js

HOW IT RUNS

`yarn install` → copy `.env.example` to `.env` → `yarn dx`

Key settings: `DATABASE_URL`, `NEXTAUTH_SECRET`, `CALENDSO_ENCRYPTION_KEY`

How a booking travels



TIME ZONES

Instants are stored in UTC and shown in each viewer's zone with Day.js.

AVAILABILITY

Slots are computed on the server from schedules, overrides, buffers and existing bookings.

OUTSIDE THE APP

Connected calendars and email are reached through packages/app-store.

Data model and API

CORE MODELS

User	timeZone, schedules
EventType	length, before/after buffer
Schedule	timeZone, owner
Availability	days, start, end, date override
Booking	uid, startTime, endTime, status
Attendee	email, timeZone

MAIN ROUTES

Get free slots

tRPC `slots.getSchedule`

Create booking

POST `/api/book/event`

Cancel booking

`/api/cancel`

Manage schedules

tRPC `availability.schedule`

List bookings

tRPC `bookings.get`

Gaps and how we found them

1

Overlap is checked in code

No database constraint blocks two overlapping bookings for one host.

2

Buffers live on the event type

They are applied when slots are computed, not stored with the booking.

3

Booking logic is spread wide

The flow crosses many packages, so behaviour is hard to trace and test.

PROCESS

Nine stages from recon to verification. Every claim carries file:line evidence. Clean room: no source copied.